Formulas:

Here's a comprehensive list of Aptitude and Reasoning formulas:

1. TIME AND WORK

Basic Formulas:

- Work = Time × Efficiency
- Time = Work / Efficiency
- Efficiency = Work / Time
- If A can do work in 'n' days, work done in 1 day = 1/n
- If A and B together: 1/A + 1/B = 1/T (where T = time together)
- If A is twice as efficient as B: A:B = 2:1, Time ratio = 1:2

Work and Wages:

- Wage ∝ Work done
- If work ratio is a:b, then wage ratio is also a:b

2. TIME, SPEED AND DISTANCE

Basic Formulas:

- Speed = Distance/Time
- Distance = Speed × Time
- Time = Distance/Speed

Conversion:

- km/hr to m/s: multiply by 5/18
- m/s to km/hr: multiply by 18/5

Average Speed:

- Average Speed = Total Distance/Total Time
- For equal distances at speeds S_1 and S_2: Average Speed = $2S_1S_2/(S_1+S_2)$

Relative Speed:

- Objects moving in same direction: Relative Speed = $S_1 - S_2$
- Objects moving in opposite direction: Relative Speed = $S_1 + S_2$

Trains:

- Time to cross a pole/person = Length of train/Speed

- Time to cross a platform = (Length of train + Length of platform)/Speed
- Two trains crossing: Time = $(L_1 + L_2)/(S_1 + S_2)$ [opposite direction]

Boats and Streams:

- Downstream speed = Speed in still water + Stream speed
- Upstream speed = Speed in still water - Stream speed
- Speed in still water = (Downstream + Upstream)/2
- Stream speed = (Downstream - Upstream)/2

3. PERCENTAGE

Basic Formulas:

- Percentage = (Value/Total) × 100
- Value = (Percentage × Total)/100
- If A is x% more than B: B is [x/(100+x)] × 100% less than A
- If A is x% less than B: B is [x/(100-x)] × 100% more than A

Population/Value Changes:

- Final value = Initial × $(1 \pm r/100)^n$ [n = number of years]

4. PROFIT AND LOSS

(Already covered in previous response - see above)

5. SIMPLE INTEREST (SI)

- SI = (P × R × T)/100
- Amount = P + SI = P(1 + RT/100)
- P = (SI × 100)/(R × T)
- R = (SI × 100)/(P × T)
- T = (SI × 100)/(P × R)

6. COMPOUND INTEREST (CI)

- Amount = $P(1 + R/100)^n$
- CI = $P(1 + R/100)^n - P$
- CI = $P[(1 + R/100)^n - 1]$

When compounded half-yearly:

- $A = P(1 + R/200)^{2n}$

When compounded quarterly:

- $A = P(1 + R/400)^{4n}$

Difference between CI and SI for 2 years:

- CI - SI = $P(R/100)^2$

7. RATIO AND PROPORTION

Basic:

- If a:b = c:d, then ad = bc
- a:b:c = x:y:z means a/x = b/y = c/z

Componendo-Dividendo:

- If a/b = c/d, then (a+b)/(a-b) = (c+d)/(c-d)

Fourth Proportional:

- If a:b = c:x, then x = bc/a

Mean Proportional:

- Mean proportional of a and b = $\sqrt{ab}$

8. AGES

- If current age ratio is a:b, after n years: (a+n):(b+n)
- n years ago: (a-n):(b-n)
- If A is x years older than B: A = B + x

9. AVERAGES

- Average = Sum of observations/Number of observations
- Sum = Average × Number of observations
- New average when one value changes: New Avg = Old Avg + (Increase or Decrease)/n

10. MIXTURES AND ALLEGATIONS

Alligation Rule:

Cheaper quantity : Dearer quantity = (d - m) : (m - c)

Where: c = cheaper price, d = dearer price, m = mean price

11. PROBABILITY

- P(E) = Number of favorable outcomes/Total outcomes
- P(not E) = 1 - P(E)
- P(A or B) = P(A) + P(B) - P(A and B)
- P(A and B) = P(A) × P(B) [for independent events]

12. PERMUTATIONS AND COMBINATIONS

Factorial:

- $n! = n \times (n-1) \times (n-2) \times ... \times 1$
- $0! = 1$

Permutations:

- ${}^nP_r = n!/(n-r)!$
- Permutation of n different things = n!
- Circular permutation = (n-1)!

Combinations:

- ${}^nC_r = n!/[r!(n-r)!]$
- ${}^nC_r = {}^nC_{n-r}$
- ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$

13. SERIES

Arithmetic Progression (AP):

- nth term: $a_n = a + (n-1)d$
- Sum: $S_n = n/2[2a + (n-1)d]$
- Sum: $S_n = n/2[a + l]$ where l = last term

Geometric Progression (GP):

- nth term: $a_n = ar^{n-1}$
- Sum of n terms: $S_n = a(r^n - 1)/(r - 1)$ when $r > 1$
- Sum of infinite GP: $S_\infty = a/(1-r)$ when $|r| < 1$

Special Series:

- Sum of first n natural numbers: $n(n+1)/2$
- Sum of squares: $n(n+1)(2n+1)/6$
- Sum of cubes: $[n(n+1)/2]^2$

14. CALENDAR

- Odd days: (Total days) mod 7
- Leap year: divisible by 4 (except century years must be divisible by 400)
- Ordinary year = 365 days = 52 weeks + 1 odd day
- Leap year = 366 days = 52 weeks + 2 odd days

15. CLOCKS

- Angle between hands = $|11m/2 - 30h|°$ (where h = hour, m = minutes)
- Minute hand moves 6° per minute

- Hour hand moves 0.5° per minute

16. AREA AND PERIMETER

Square:

- Area = a^2
- Perimeter = 4a
- Diagonal = $a\sqrt{2}$

Rectangle:

- Area = $l \times b$
- Perimeter = $2(l + b)$
- Diagonal = $\sqrt{(l^2 + b^2)}$

Circle:

- Area = πr^2
- Circumference = $2\pi r$
- Area of sector = $(\theta/360) \times \pi r^2$

Triangle:

- Area = $1/2 \times$ base $\times$ height
- Area = $\sqrt{[s(s-a)(s-b)(s-c)]}$ (Heron's formula) where $s = (a+b+c)/2$

17. VOLUME AND SURFACE AREA

Cube:

- Volume = a^3
- Total Surface Area = $6a^2$

Cuboid:

- Volume = $l \times b \times h$
- Total Surface Area = $2(lb + bh + hl)$

Cylinder:

- Volume = $\pi r^2 h$
- Curved Surface Area = $2\pi rh$
- Total Surface Area = $2\pi r(r + h)$

Sphere:

- Volume = $4/3\pi r^3$
- Surface Area = $4\pi r^2$

Cone:

- Volume = $1/3\pi r^2 h$
- Curved Surface Area = πrl (l = slant height)
- Total Surface Area = $\pi r(r + l)$

18. PARTNERSHIP

- Profit ratio = Capital ratio × Time ratio
- If A invests for x months and B for y months: Profit ratio = $(A \times x):(B \times y)$

19. PIPES AND CISTERNS

- If pipe fills in x hours: Rate = $1/x$ per hour
- If pipe empties in y hours: Rate = $-1/y$ per hour
- Net rate = Sum of individual rates

20. BLOOD RELATIONS

- Remember generation gaps
- Mother's or Father's side relationships
- Direct relations vs. indirect relations

These formulas cover most aptitude and reasoning topics for competitive exams!